

TPS65219 PMIC EVM

Power-Rail Measurement Report

Generated : 07 May 2026 11:14:15

Instrument : Keysight EDUX1052A | Timebase : 200 ms/div | DC Tol : $\pm 1V$ | BUCK : tRAMP (10%→98%) | LDO : Slew (10%→90%)

Total Events	Rails Tested	Snapshots
19	7	19

Rail Configuration

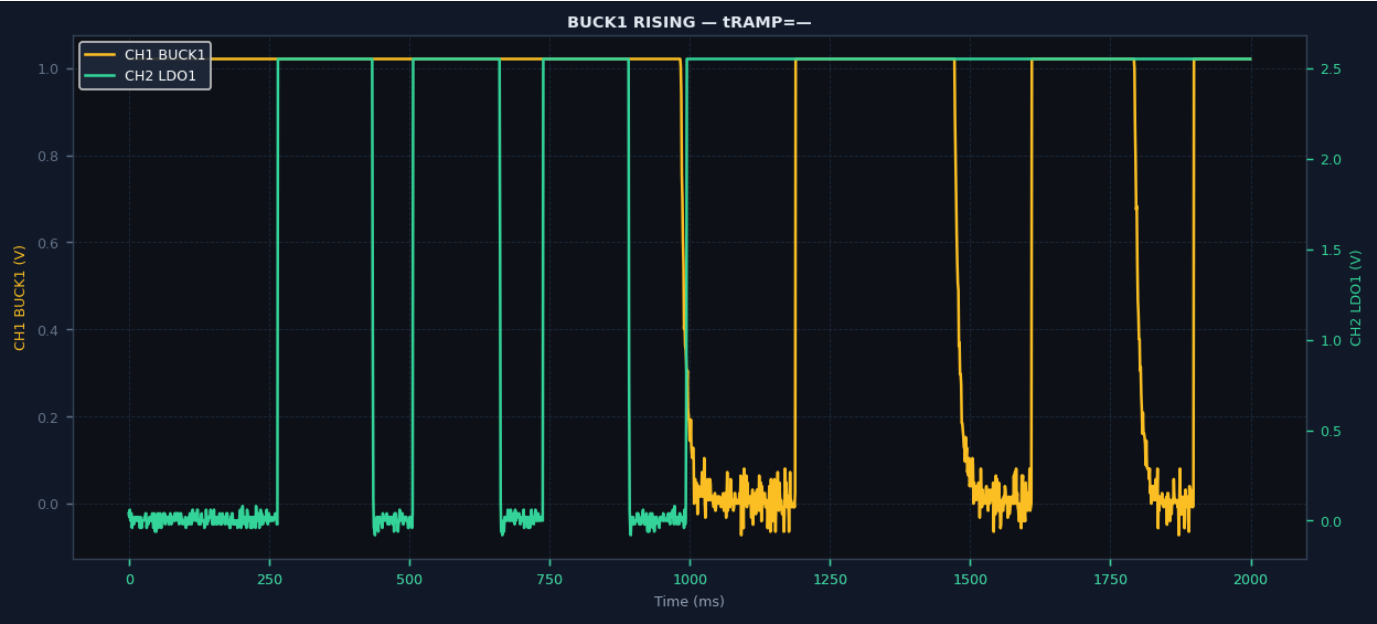
Rail	Type	VNOM (V)	V/div	Trig V	Spec Param	Spec Min	Spec Max	Unit
BUCK1	Buck DC-DC	0.749	0.20	0.10	ms tRAMP	1.0	500.0	ms
BUCK2	Buck DC-DC	3.289	1.00	0.30	ms tRAMP	1.0	500.0	ms
BUCK3	Buck DC-DC	1.197	0.50	0.10	ms tRAMP	1.0	500.0	ms
LDO1	LDO type 1	1.796	0.50	0.20	mV/μs Slew	10.0	14.0	mV/μs
LDO2	LDO type 1	0.848	0.20	0.10	mV/μs Slew	10.0	14.0	mV/μs
LDO3	LDO type 2	1.796	0.50	0.20	mV/μs Slew	7.5	27.0	mV/μs
LDO4	LDO type 2	2.496	1.00	0.20	mV/μs Slew	7.5	27.0	mV/μs

Waveform Snapshots & Results

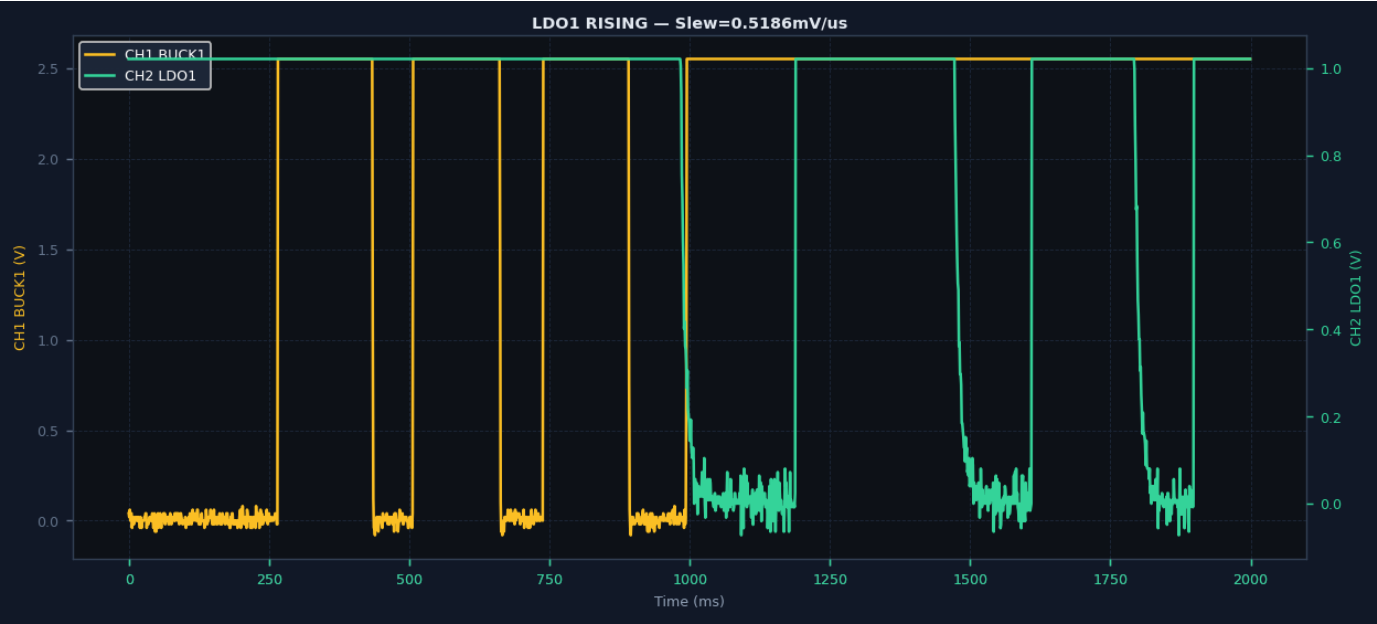
Snapshot 1 | Rail: LDO1 | CH: CH2 | Direction: FALLING | VNOM=1.796V | Result: **PASS**



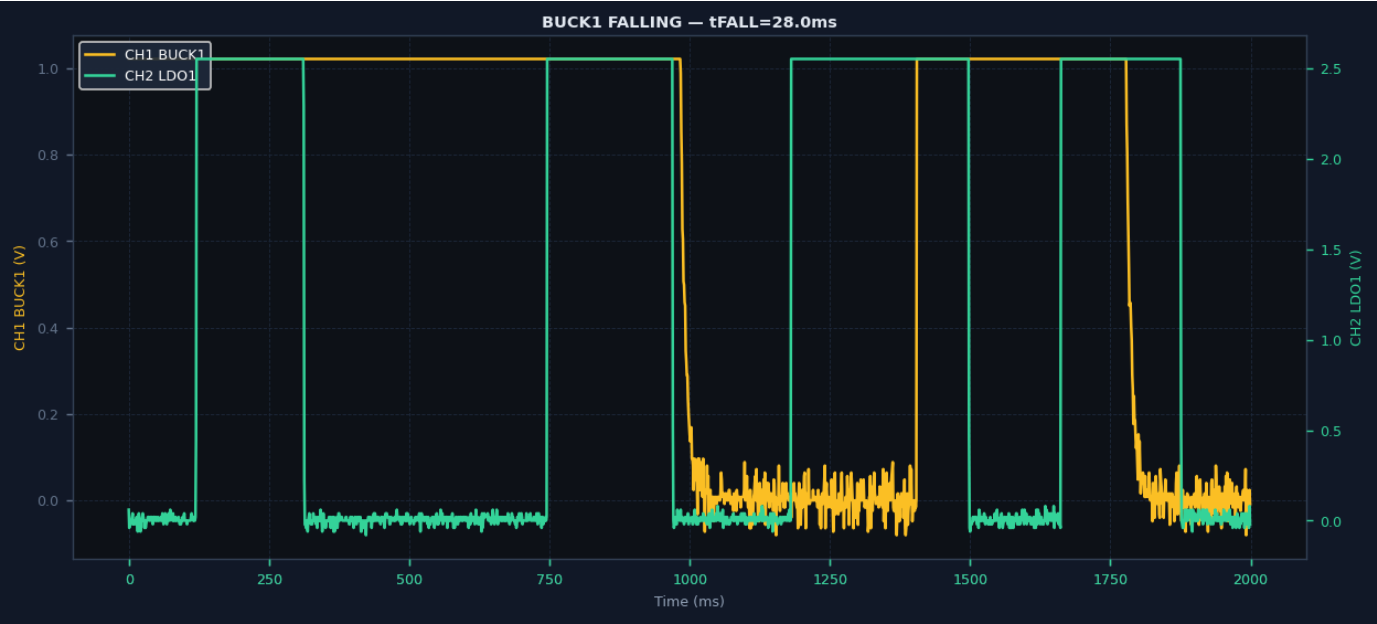
Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.4136mV/us	10.0–14.0mV/us	PASS
DC Voltage	1.293V	0.796–2.796V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=-	1.0-500.0ms	PASS
DC Voltage	0.546V	-0.251-1.749V (±1V)	PASS



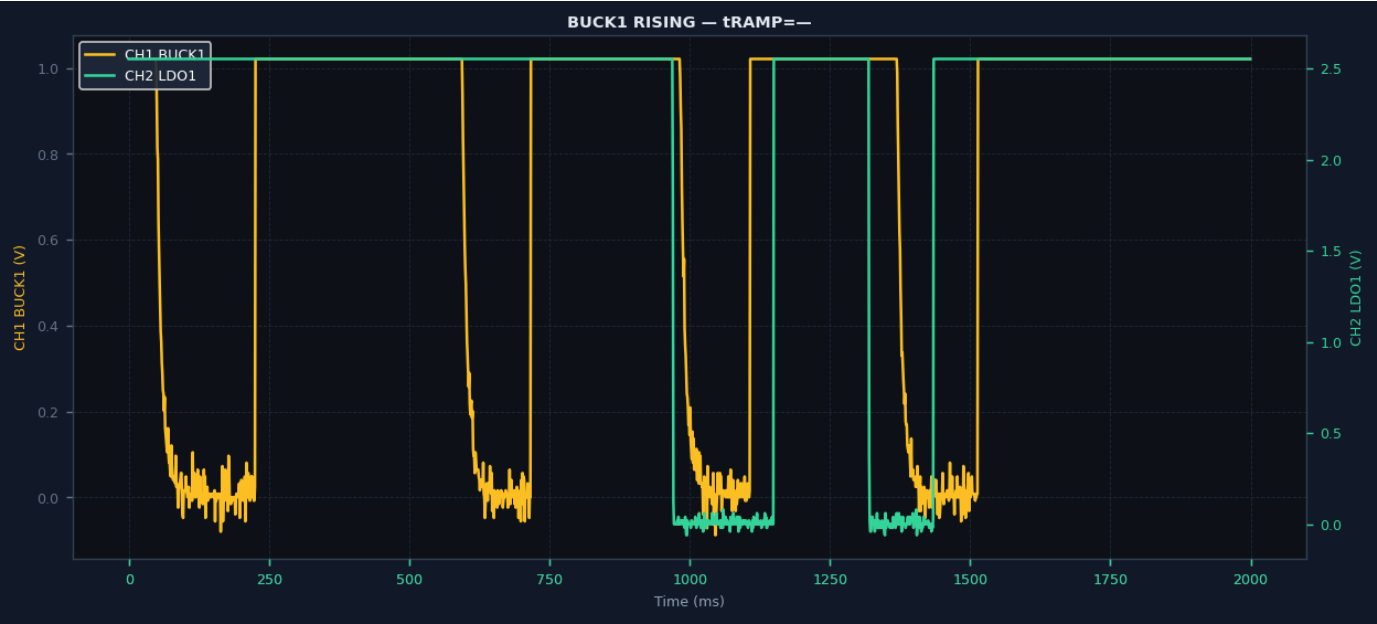
Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.5186mV/us	10.0–14.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V (±1V)	PASS



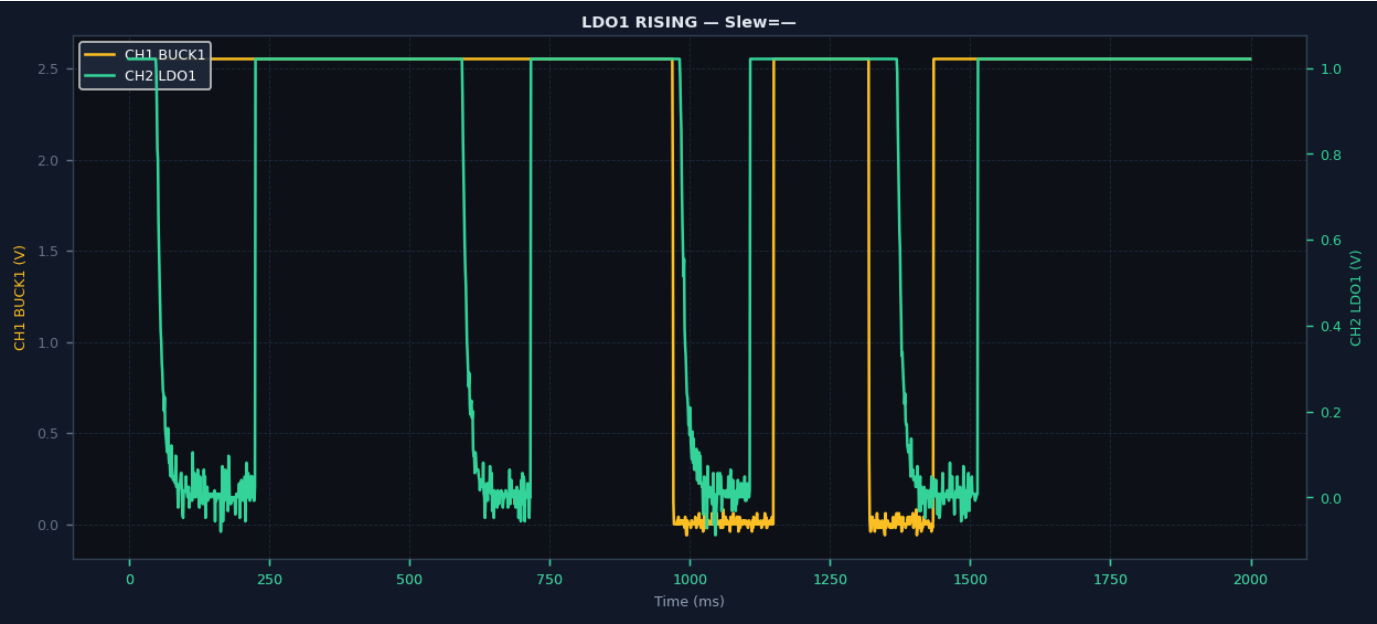
Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=28.0ms	—	PASS
DC Voltage	0.689V	-0.251–1.749V (±1V)	PASS



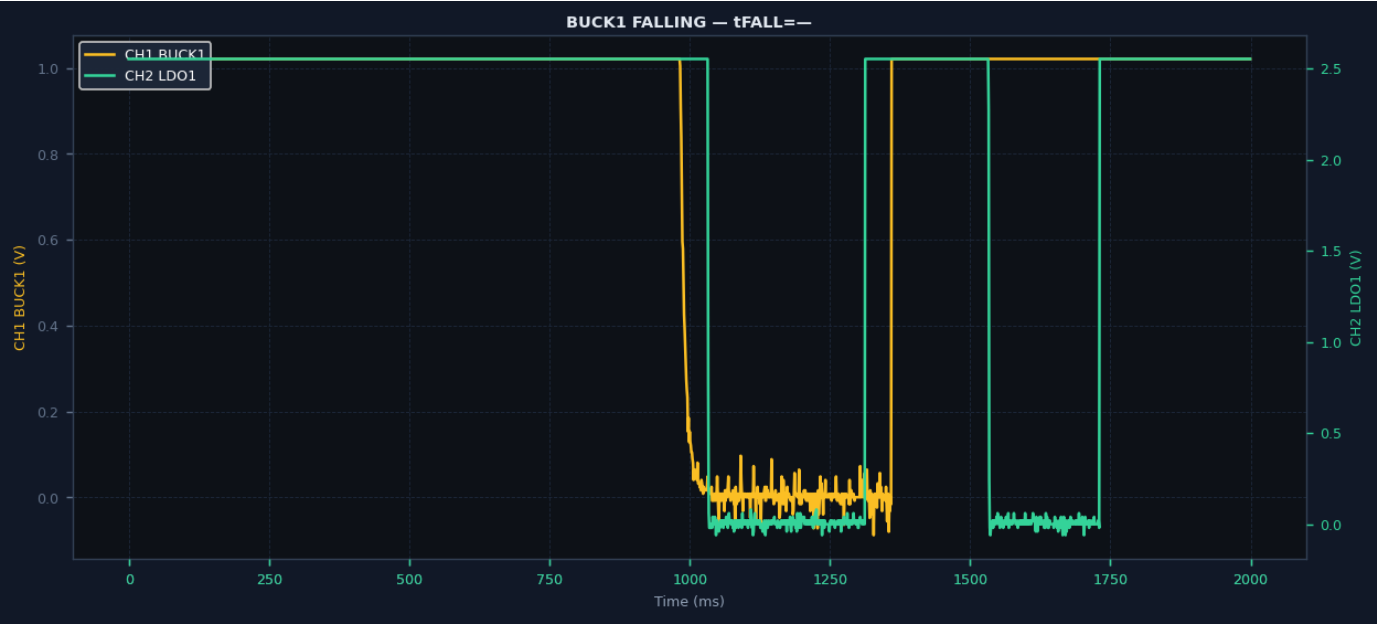
Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	10.0–14.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=—	1.0–500.0ms	PASS
DC Voltage	0.751V	-0.251–1.749V (±1V)	PASS



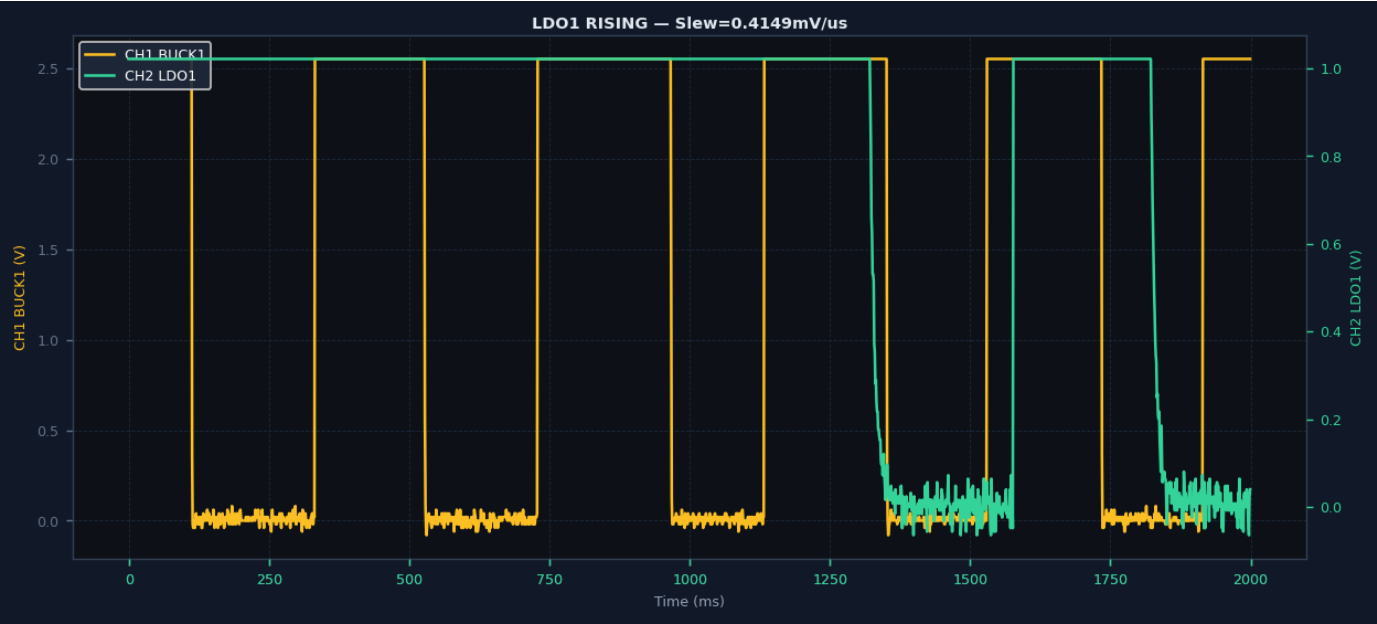
Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	10.0–14.0mV/us	PASS
DC Voltage	1.293V	0.796–2.796V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=—	—	PASS
DC Voltage	0.539V	-0.251–1.749V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=1.0291mV/us	10.0–14.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V (±1V)	PASS



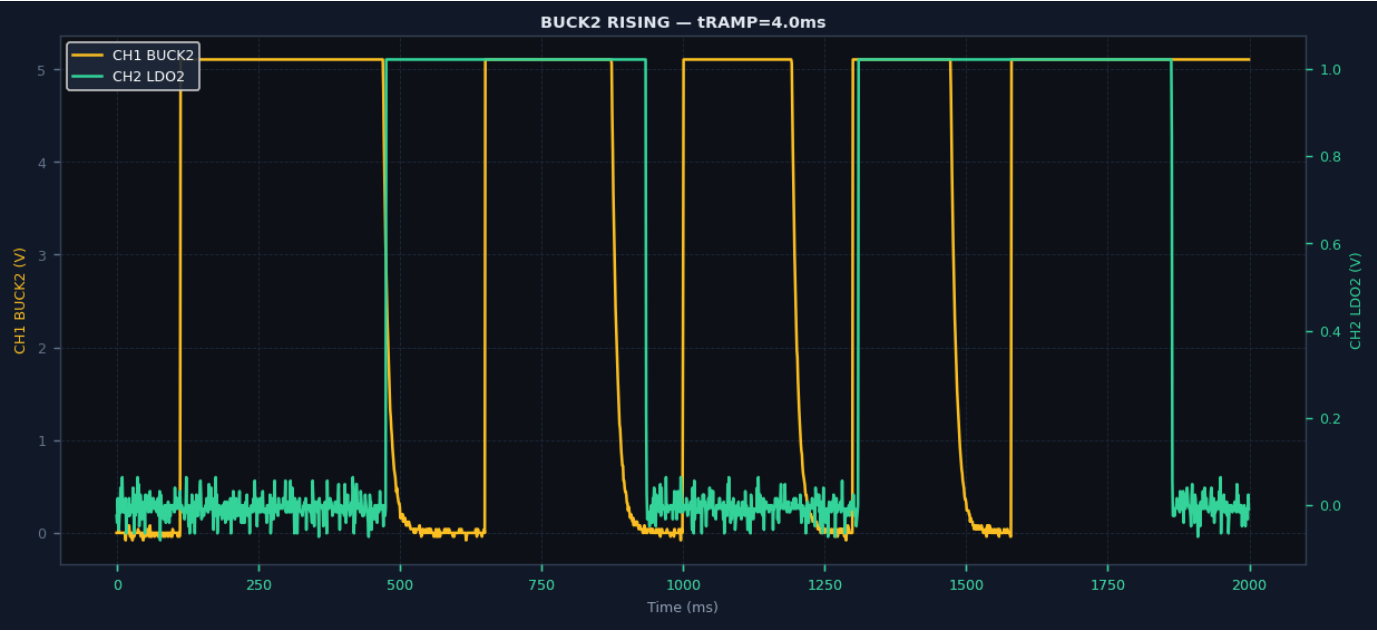
Parameter	Measured	Spec/Reference	Result
Rail	LDO1	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.4149mV/us	10.0–14.0mV/us	PASS
DC Voltage	2.553V	0.796–2.796V (±1V)	PASS



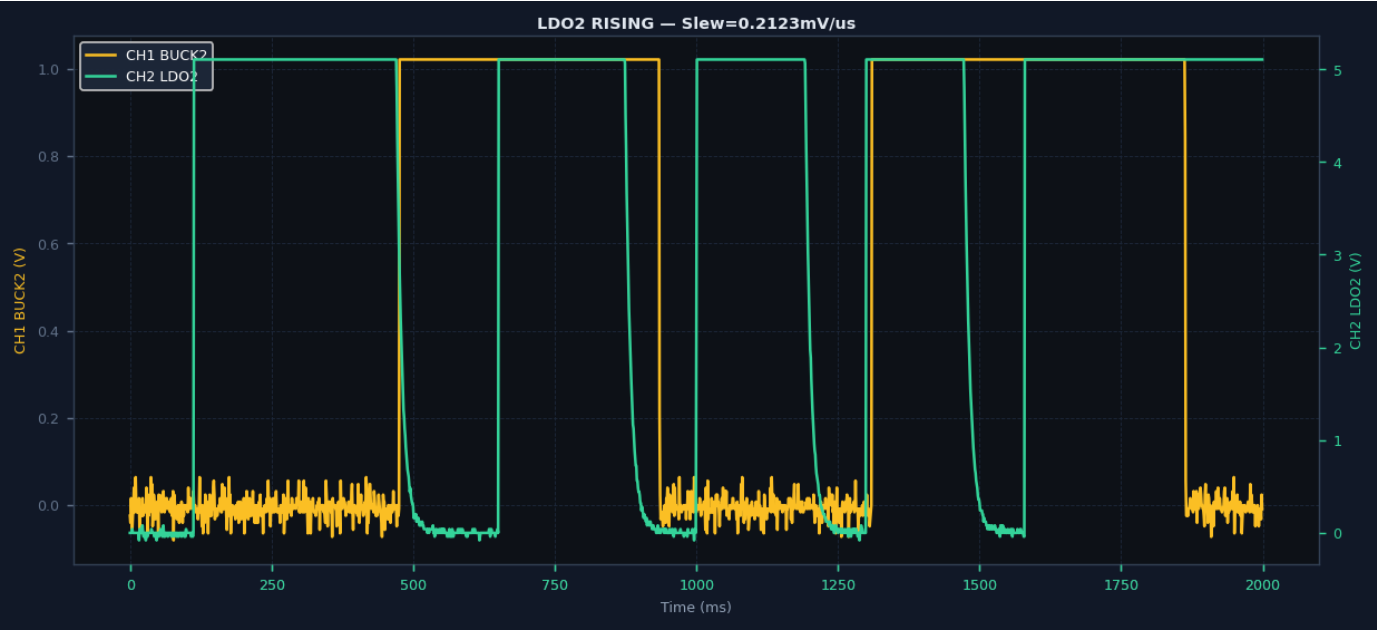
Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP==	1.0–500.0ms	PASS
DC Voltage	0.724V	-0.251–1.749V (±1V)	PASS



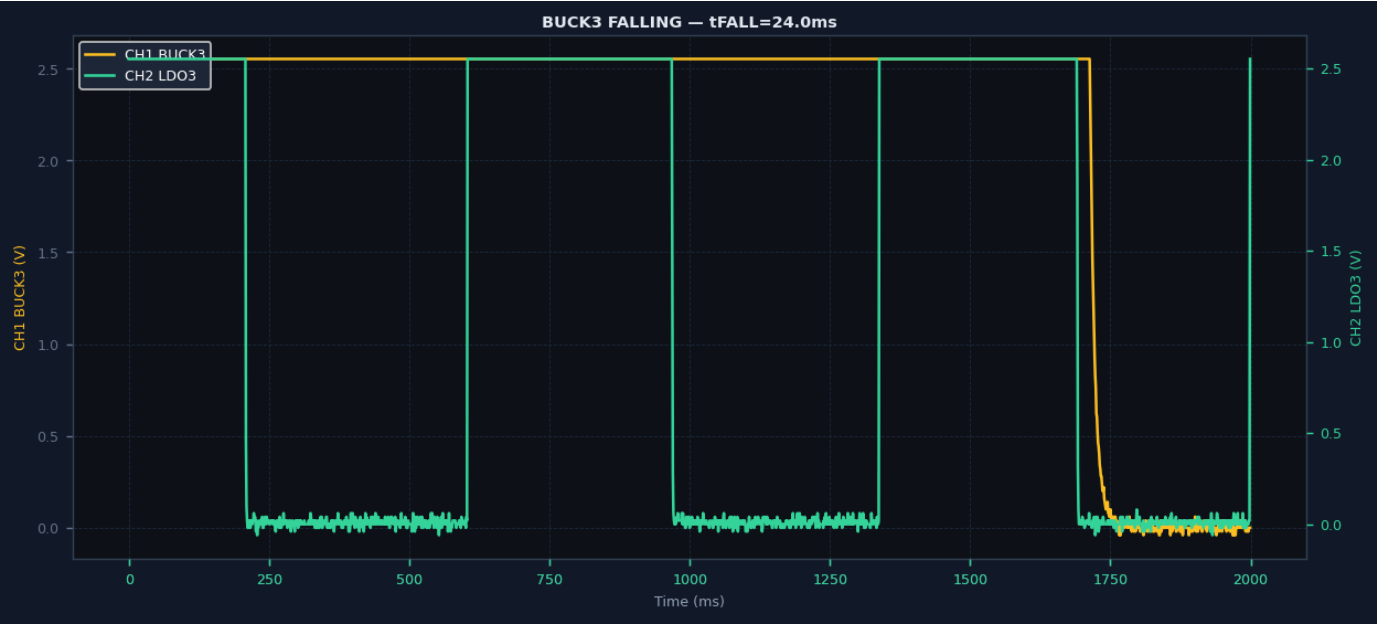
Parameter	Measured	Spec/Reference	Result
Rail	LDO2	VNOM=0.848V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	10.0–14.0mV/us	PASS
DC Voltage	0.668V	-0.152–1.848V (±1V)	PASS



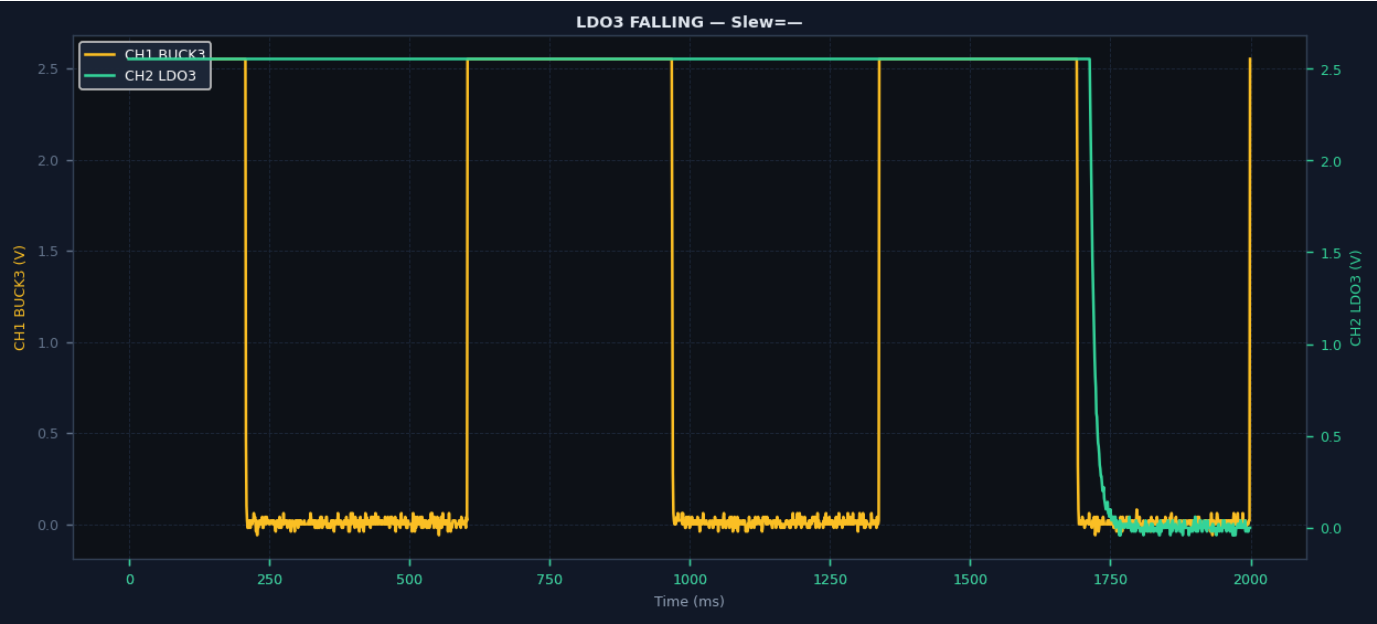
Parameter	Measured	Spec/Reference	Result
Rail	BUCK2	VNOM=3.289V	--
Direction	RISING	--	--
Method	10%→98% swing	--	--
Measurement	tRAMP=4.0ms	1.0–500.0ms	PASS
DC Voltage	3.702V	2.289–4.289V (±1V)	PASS



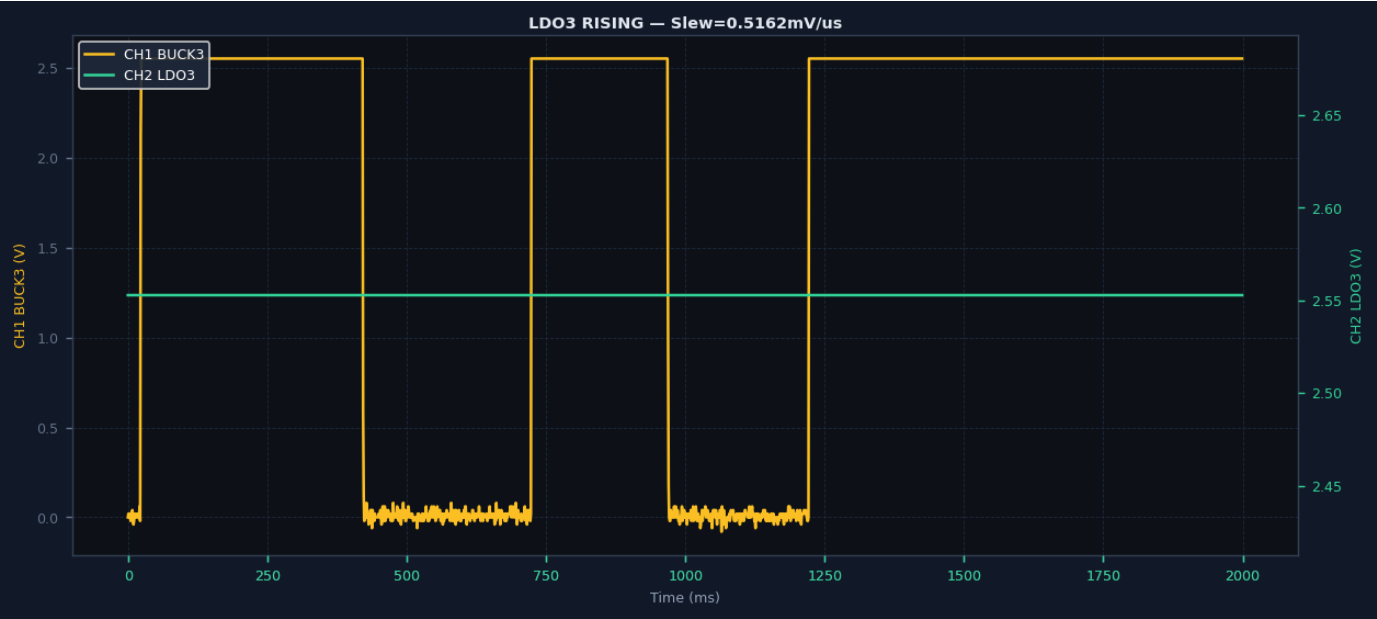
Parameter	Measured	Spec/Reference	Result
Rail	LDO2	VNOM=0.848V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.2123mV/us	10.0–14.0mV/us	PASS
DC Voltage	0.507V	-0.152–1.848V (±1V)	PASS



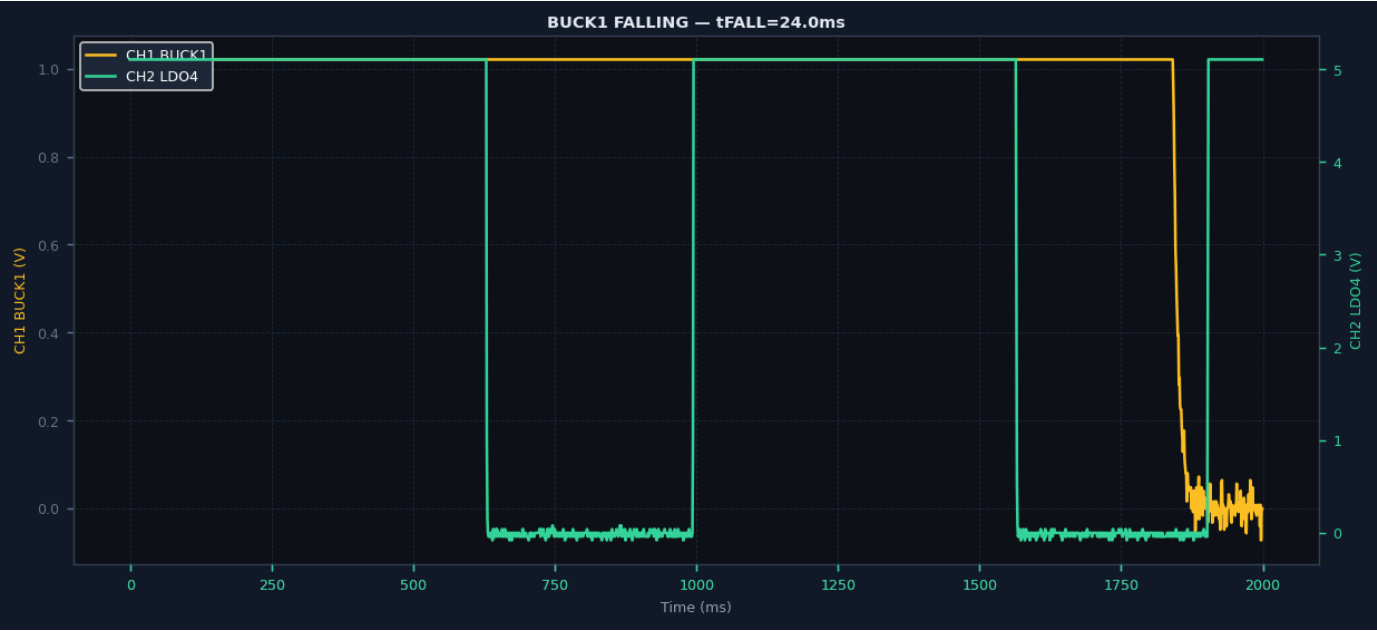
Parameter	Measured	Spec/Reference	Result
Rail	BUCK3	VNOM=1.197V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=24.0ms	—	PASS
DC Voltage	2.553V	0.197–2.197V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	FALLING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=—	7.5–27.0mV/us	PASS
DC Voltage	1.236V	0.796–2.796V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO3	VNOM=1.796V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=0.5162mV/us	7.5–27.0mV/us	PASS
DC Voltage	1.726V	0.796–2.796V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	BUCK1	VNOM=0.749V	--
Direction	FALLING	--	--
Method	98%→10% swing	--	--
Measurement	tFALL=24.0ms	—	PASS
DC Voltage	1.021V	-0.251–1.749V (±1V)	PASS



Parameter	Measured	Spec/Reference	Result
Rail	LDO4	VNOM=2.496V	--
Direction	RISING	--	--
Method	10%→90% swing	--	--
Measurement	Slew=1.0308mV/us	7.5–27.0mV/us	PASS
DC Voltage	5.106V	1.496–3.496V (±1V)	PASS

Complete Measurement Log

#	Rail	Ch	Dir	Measured	Spec Range	DC (V)	P/F
1	LDO1	CH2	FALLING	Slew=0.4136mV/us	10.0–14.0mV/us	1.293V	PASS
2	BUCK1	CH1	RISING	tRAMP=—	1.0–500.0ms	0.546V	PASS
3	LDO1	CH2	RISING	Slew=0.5186mV/us	10.0–14.0mV/us	2.553V	PASS
4	BUCK1	CH1	FALLING	tFALL=28.0ms	—	0.689V	PASS
5	LDO1	CH2	FALLING	Slew=—	10.0–14.0mV/us	2.553V	PASS
6	BUCK1	CH1	RISING	tRAMP=—	1.0–500.0ms	0.751V	PASS
7	LDO1	CH2	RISING	Slew=—	10.0–14.0mV/us	1.293V	PASS
8	BUCK1	CH1	FALLING	tFALL=—	—	0.539V	PASS
9	LDO1	CH2	FALLING	Slew=1.0291mV/us	10.0–14.0mV/us	2.553V	PASS
10	LDO1	CH2	RISING	Slew=0.4149mV/us	10.0–14.0mV/us	2.553V	PASS
11	BUCK1	CH1	RISING	tRAMP=—	1.0–500.0ms	0.724V	PASS
12	LDO2	CH2	FALLING	Slew=—	10.0–14.0mV/us	0.668V	PASS
13	BUCK2	CH1	RISING	tRAMP=4.0ms	1.0–500.0ms	3.702V	PASS
14	LDO2	CH2	RISING	Slew=0.2123mV/us	10.0–14.0mV/us	0.507V	PASS
15	BUCK3	CH1	FALLING	tFALL=24.0ms	—	2.553V	PASS
16	LDO3	CH2	FALLING	Slew=—	7.5–27.0mV/us	1.236V	PASS
17	LDO3	CH2	RISING	Slew=0.5162mV/us	7.5–27.0mV/us	1.726V	PASS
18	BUCK1	CH1	FALLING	tFALL=24.0ms	—	1.021V	PASS
19	LDO4	CH2	RISING	Slew=1.0308mV/us	7.5–27.0mV/us	5.106V	PASS

Specification Reference

Rail	Parameter	Method	Min	Max	Unit
BUCK1/2/3	tRAMP	10%→98% swing	1.0	500.0	ms (EVM)
BUCK IC	tRAMP	IC datasheet	0.30	1.65	ms
LDO1/LDO2	Slew	10%→90% swing	10	14	mV/μs
LDO3/LDO4	Slew	10%→90% swing	7.5	27	mV/μs

DC voltage tolerance: ±1V from VNOM. All waveform measurements at 200ms/div on Keysight EDUX1052A.